

CLOUDFLARE RADAR

# 2024 Year in Review

Internet patterns and trends observed through Cloudflare's global network during 2024

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Audience: Technology Executives & Network Infrastructure Leaders

Data Source: Cloudflare's global network — millions of sites & applications

Methodology: Aggregate traffic, DNS (1.1.1.1), and security telemetry

# 17% Global Traffic Growth and Starlink's 3.3x Surge

Global Internet traffic trends & satellite connectivity expansion

**+17%**

YoY Global Traffic Growth

**3.3×**

Starlink Traffic Surge

**41%**

Mobile Device Traffic

**28%**

IPv6 Dual-Stack Adoption

Global Internet traffic grew 17% year-over-year, driven by rising reliance on communication, commerce, entertainment, and transportation services.

SpaceX Starlink traffic surged 3.3× during 2024, expanding satellite Internet connectivity to previously underserved and remote regions worldwide.

Trend baseline starts mid-January to normalize for post-holiday activity spikes.

41% of global traffic originated from mobile devices.

# Google Leads Top 10 Services as Generative AI Rises

Most popular Internet services — based on Cloudflare 1.1.1.1 DNS resolver data

| Rank | Service   | Category             |
|------|-----------|----------------------|
| #1   | Google    | Search & Cloud       |
| #2   | Facebook  | Social Media         |
| #3   | Apple     | Devices & Services   |
| #4   | TikTok    | Short-form Video     |
| #5   | AWS       | Cloud Infrastructure |
| #6   | Microsoft | Productivity & Cloud |
| #7   | Instagram | Social / Photo       |
| #8   | YouTube   | Video Streaming      |
| #9   | Amazon    | E-commerce           |
| #10  | WhatsApp  | Messaging            |

## GENERATIVE AI MOMENTUM

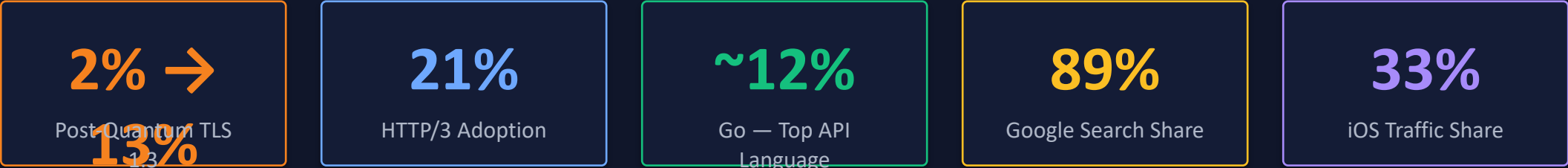
Generative AI continued its meteoric rise throughout 2024, reshaping how users interact with Internet services. Meanwhile, the Metaverse receded into the background.

## Key Methodology Insights

Rankings derived from aggregate 1.1.1.1 DNS resolver data. Google held #1 for the second consecutive year. Social platforms occupy 4 of the top 10 positions. Cloud infrastructure (AWS, Microsoft) remains dominant.

# Post-Quantum Encryption Jumps from 2% to 13% of TLS 1.3

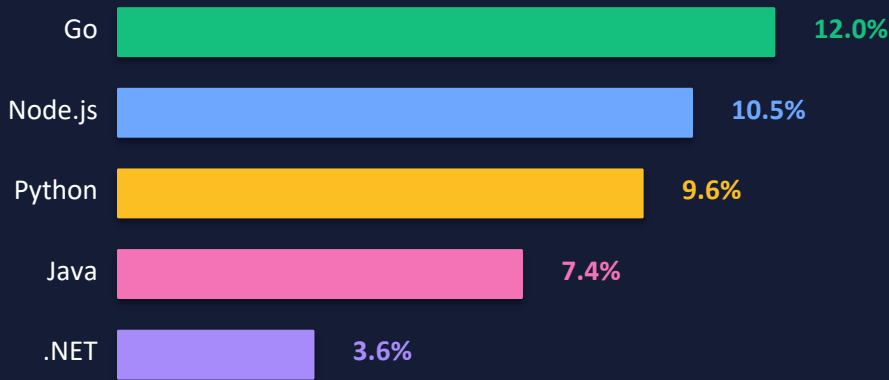
Technology adoption shifts: encryption, protocols, and developer ecosystem



## Post-Quantum Encryption Adoption

Started at ~2% of TLS 1.3 traffic in early 2024.  
Chrome 124 enabled PQ key agreement by default (Apr 17).  
Adoption rapidly climbed to ~12-13% by year-end.  
Mozilla Firefox began rolling out PQ by default.  
Apple Safari started initial PQ testing.

## API Client Language Share



## Additional Technology Insights

HTTP/3 (QUIC-based) reached 21% of all traffic.  
Chrome remains the most popular browser globally.  
Peak Android share exceeds 95% in some regions.  
Search: Google 89%, Yandex 3.1%, Baidu 2.7%, Bing 2.6%.

# 225 Major Disruptions: Government Shutdowns Lead Outage Causes

Internet connectivity disruptions tracked by Cloudflare Radar Outage Center

225

Major Disruptions in 2024

116+

Government-Directed

41

Cable Cuts

16

Power Outages

| Cause                           | Count |
|---------------------------------|-------|
| <div></div> Government Directed | 116+  |
| <div></div> Cable Cut           | 41    |
| <div></div> Unknown             | 16    |
| <div></div> Power Outage        | 16    |
| <div></div> Technical Problem   | 9     |
| <div></div> Military Action     | 9     |
| <div></div> Weather             | 8     |
| <div></div> Maintenance         | 4     |
| <div></div> Cyber Attack        | 4     |
| <div></div> Natural Disaster    | 1     |
| <div></div> Fire                | 1     |

# 6.5% of Traffic Mitigated: Persistent Threats and Bot Concentration

Security findings: mitigation, attacks, vulnerabilities, and email threats



## Traffic Mitigation & Attack Landscape

Cloudflare mitigated 6.5% of all processed traffic.  
Gambling & Games sector — most attacked industry.  
68% of global bot traffic came from just 10 countries.

## Log4j Persistence (CVE-2021-44228)

First disclosed December 2021, still actively targeted 3+ years later.  
Demonstrates persistent nature of widely-known vulnerabilities.

## Routing Security

RPKI valid routes increased 6.4% in 2024.  
Improving BGP routing security across the Internet.

## Email Security Threats

4.3% of all emails processed were malicious.  
43% of malicious emails contained deceptive links.  
.bar TLD originated largest share of malicious & spam email.

## Most Attacked Sector

**Gambling & Games — #1 Most Attacked Sector**

## Security Summary

Threats remain pervasive across all layers of the stack —  
from application exploits (Log4j) to email phishing campaigns.

# Five Takeaways Shaping the 2025 Internet Outlook

Key findings from 2024 to inform strategic planning for network and security leaders

01

## Sustained Traffic Growth Trajectory

Global Internet traffic grew 17% YoY. Starlink's 3.3× surge signals satellite connectivity will be a major growth vector, bringing underserved regions online.

02

## Post-Quantum Encryption Momentum

PQ encryption leapt from ~2% to 13% of TLS 1.3 traffic after Chrome 124's default enablement. Browser-driven adoption will accelerate — plan for PQ readiness now.

03

## Connectivity Fragility from Government Actions

Government-directed shutdowns led 225 major disruptions (116+ incidents). Cable cuts (41) and power outages (16) round out the top causes.

04

## Persistent & Evolving Security Threats

6.5% of all traffic mitigated. Log4j exploitation persists 3+ years post-disclosure. Gambling & Games most targeted; 68% of bot traffic from just 10 countries.

05

## Developer Ecosystem Evolution

Go surpassed all languages as the top API client (~12% of automated traffic). HTTP/3 reached 21%. The developer toolchain continues to shift toward performance-first choices.